ZIYU WANG

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EDUCATION

XI'AN JIAOTONG UNIVERSITY **Xi'an, China**
Bachelor of Science in Physics Sept. 2020 - June 2024

- **Average Grade:** 90.03/100; **GPA:** 3.83/4.3
- **Courses:** Mechanics (100/100), Electromagnetism (90/100), General Thermal Physics (96/100), Optics (91/100), General Physics-Atomic Physics (99/100), Theoretical Mechanics (98/100), Electrodynamics (94/100), Mathematical Methods in Physics (95/100)
- **Honors:**
  - **Finalist Prize** of COMAP's Mathematical Contest in Modeling (MCM)® (Top 2% of all participating teams worldwide) (2022)
  - School First Prize in Xi'an Jiaotong University (2022)
  - Outstanding student in Xi'an Jiaotong University (2022)
  - 2022 Provincial-level College Student Innovation Training Project - Excellent (2022)
  - 33rd Tengfei Cup School-level Excellence Award (2022)
  - Provincial First Prize in National Mathematics Contest (2021)

UNIVERSITY OF CALIFORNIA, BERKELEY **Berkeley, CA**
Exchange Student Jan. 2023 – Aug. 2023

- **GPA:** 4.00/4.00
- **Courses:** Quantum Mechanics (A+), Special Relativity and General Relativity (A+), Solid State Physics (A+)

PUBLICATIONS AND PATENTS

- Z. Zhou, **Z. Wang**, Z. Tang, et al. (2023). A plant-wearable self-powering system based on droplet frictional electricity and its method. China Patent Application No. CN115967297A. China National Intellectual Property Administration.
- R. Qi, A. Y. Joe, … **Z. Wang**, … F. Wang, (submitted) Perfect Coulomb drag and exciton transport in an excitonic insulator.

RESEARCH EXPERIENCE

RESEARCH ON MOIRÉ EXCITONS IN $MoSe_2/WS_2$ HETEROSTRUCTURES **Berkeley, CA**
*Advisor: Prof. Feng Wang, Department of Physics, **UC Berkeley*** Jan. 2023 – Aug. 2023

- Created devices of 2D materials and conducted optical measurements (absorption spectroscopy and photoluminescence measurements) on 2D devices and analyzed experimental data.
- Calculated the reflectance spectrum of given 2D devices using the transfer-matrix method.
- Calculated the dipole-dipole interaction and exchange of excitons on moiré superlattices.

BOGOLIUBOV QUASIPARTICLE INTERFERENCE IMAGING OF UNCONVENTIONAL SUPERCONDUCTORS **Notre Dame, IN**
*Advisor: Prof. Xiaolong Liu, Department of Physics & Astronomy, **Notre Dame University*** May 2022 – Aug. 2022

- Developed a MATLAB GUI to calculate quasiparticle interference patterns for different materials.
- Calculated quasiparticle interference patterns of $Bi_2Sr_2CaCu_2O_{8+\delta}$ by giving a predicted energy band and energy gap structure and compared calculated results with experimental observations.

RESEARCH ON SUPERHYDROPHOBIC TRIBOELECTRIC NANOGENERATORS **Xi'an, China**
*Advisor: Prof. Zhifu Zhou, School of Energy and Power Engineering, **Xi'an Jiaotong University*** Sept. 2021 - May 2022

- Improved the performance of flexible triboelectric nanogenerators through surface treatment technology and Laser-Induced Graphene technology.
- Designed a Rainwater Energy Harvester using flexible triboelectric nanogenerators and piezoelectric materials.
- Wrote a national patent as the first student-author.

ACTIVITIES

XI'AN JIAOTONG UNIVERSITY ABU-ROBOCON ROBOTICS TEAM **Xi'an, China**
A Member of Software Programming Group Aug. 2021 – Nov. 2021

- Wrote and uploaded programs to STM32 microcontrollers using Keil and J-Link software

RESEARCH ON THE PRIMARY AND SECONDARY SCHOOL ENROLLMENT SYSTEM IN XI'AN **Xi'an, China**
Social Practice Activity May 2021 – Aug. 2021

- Distributed questionnaires and conducted interviews with parents of primary and secondary school students, analyzed existing educational policies, and presented suggestions for improvement to the Education Bureau of Xi'an City

SKILLS

- **Technical skills:** MATLAB, LaTeX, Mathematica, Python, C++, SQL
- **Languages:** Mandarin (native), English (fluent)